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(54) SYSTEMS AND METHODS OF LIDAR SENSOR SYSTEMS HAVING AMPLIFIER PROTECTION CIRCUITS

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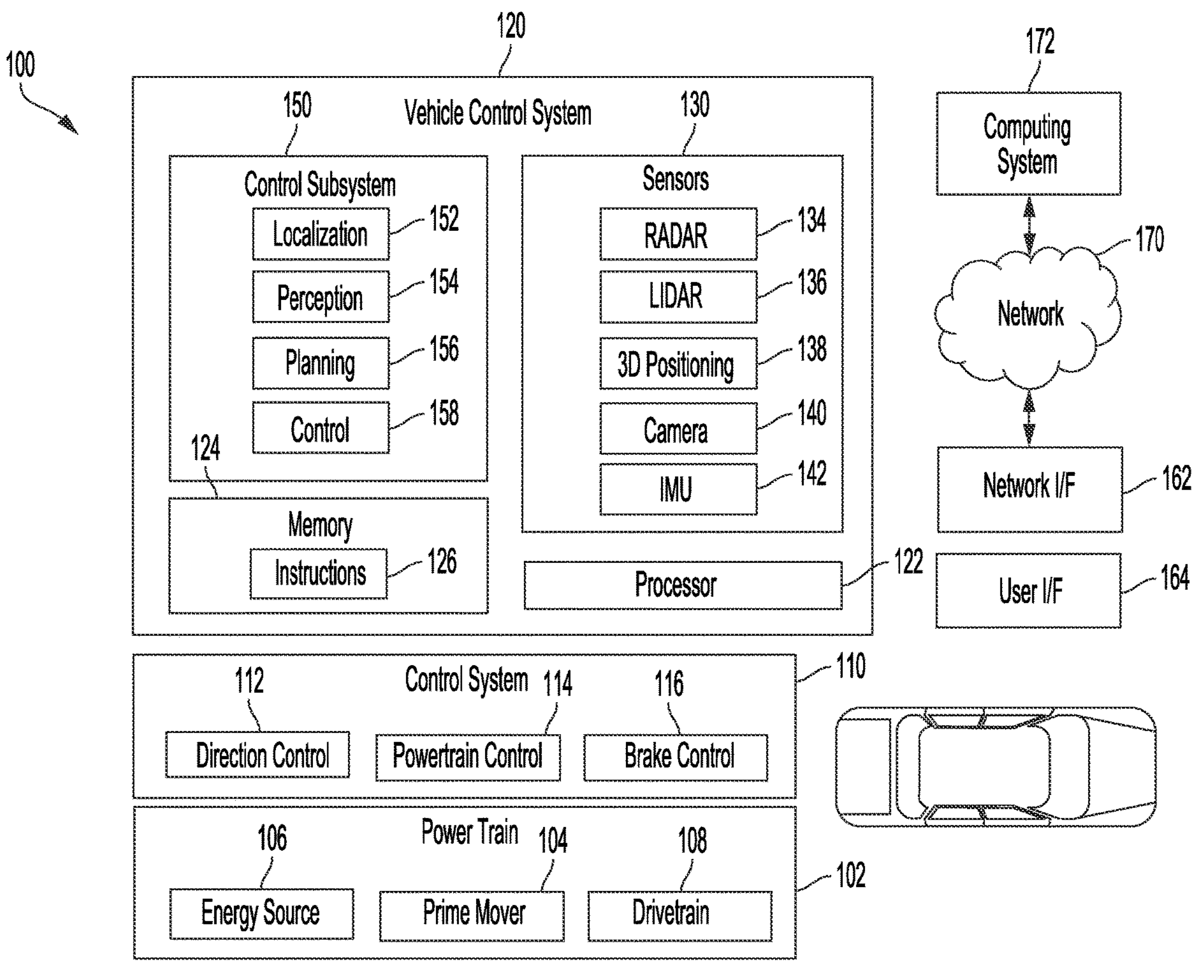
(57) **ABSTRACT**

A light detection and ranging (LIDAR) sensor system includes a laser source, an amplifier, a circuit coupled with the amplifier, and one or more scanning optics. The laser source is configured to output a beam. The amplifier is configured to amplify the beam to provide an amplified beam. The circuit is configured to provide power to the amplifier to cause the amplifier to amplify the beam. The circuit is configured to control a parameter of the power. The one or more scanning optics are configured to receive the amplified beam from the amplifier and output the amplified beam.



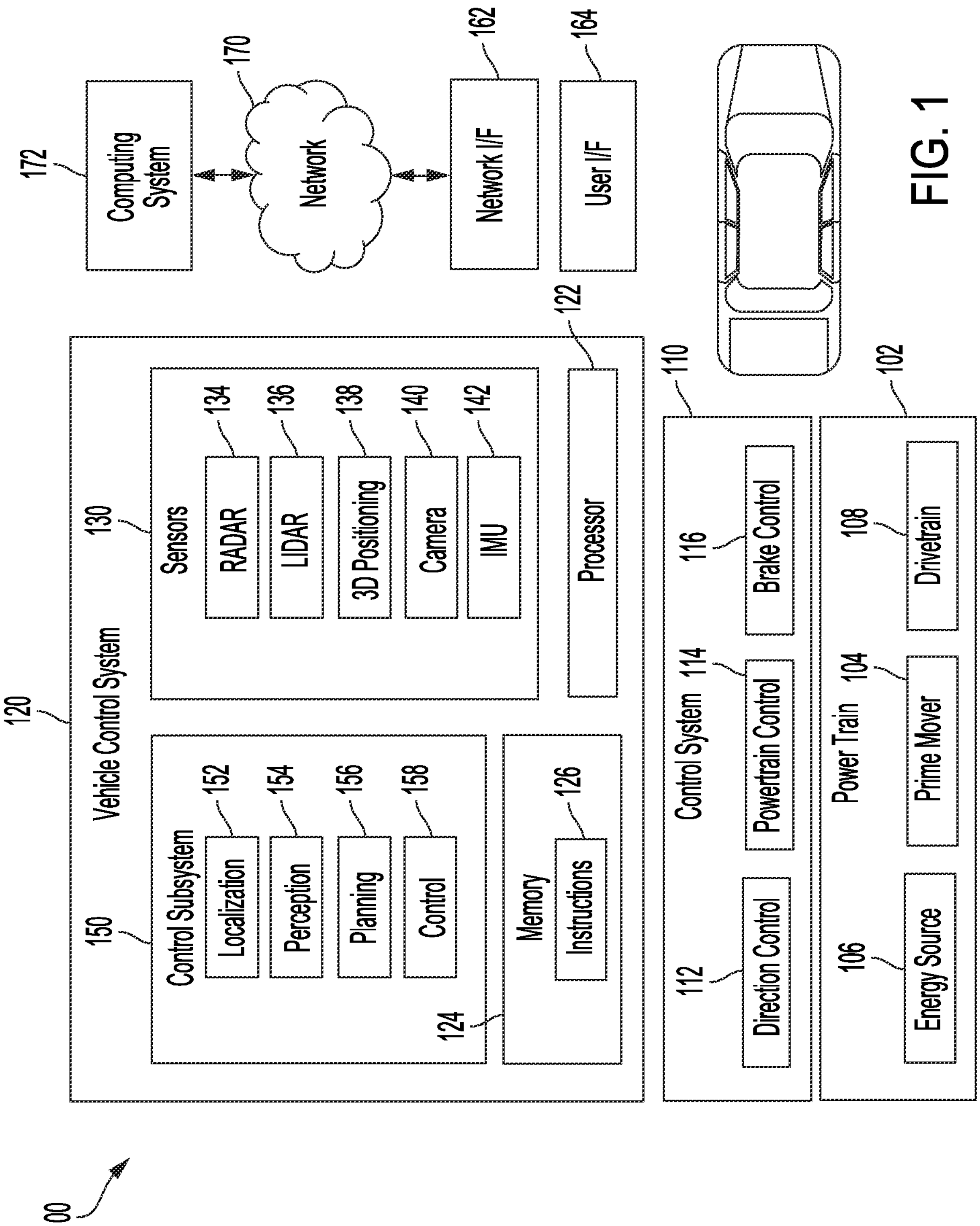

100
120
Vehicle Control System
150
Control Subsystem
152
Localization
154
Perception
156
Planning
158
Control
124
Memory
126
Instructions
130
Sensors
134
RADAR
136
LIDAR
138
3D Positioning
140
Camera
142
IMU
122
Processor
110
Control System
112
Direction Control
114
Powertrain Control
116
Brake Control
102
Power Train
106
Energy Source
104
Prime Mover
108
Drivetrain
172
Computing System
170
Network
162
Network I/F
164
User I/F

FIG. 1

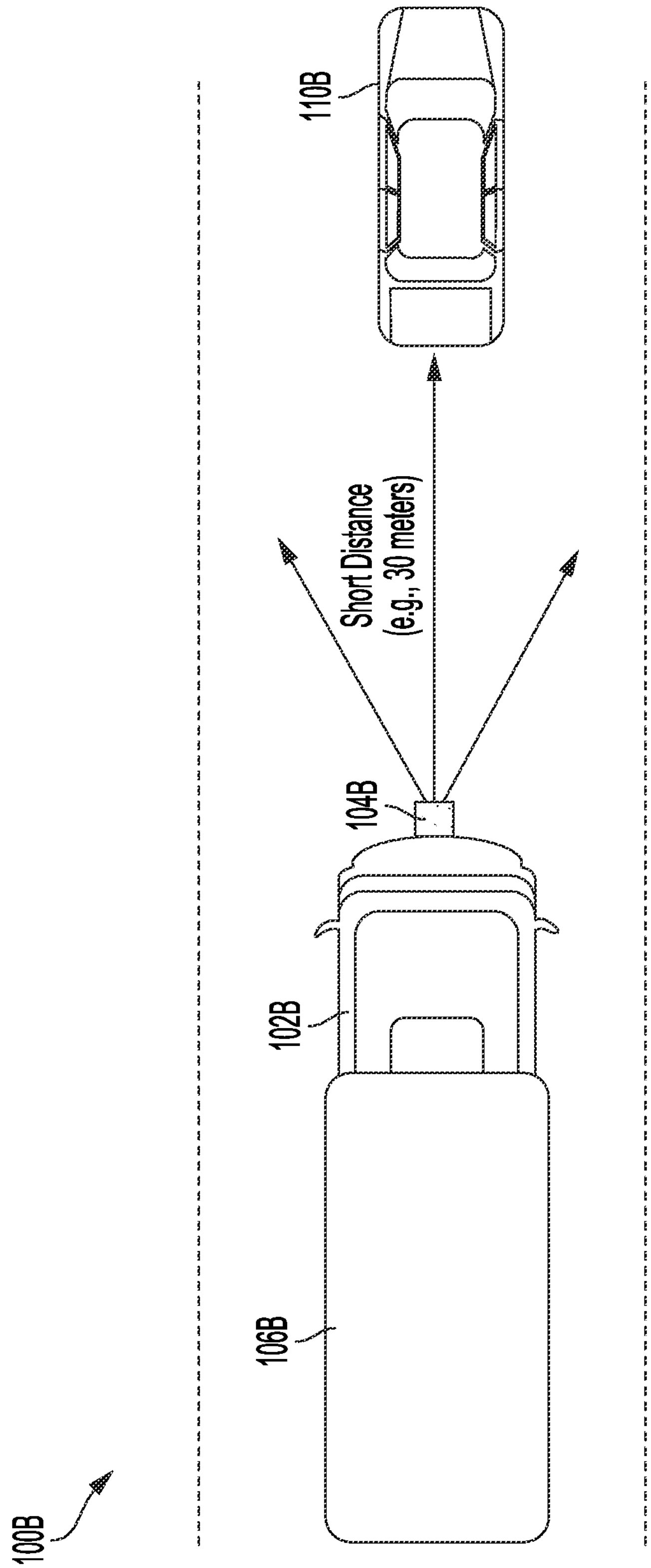


FIG. 2

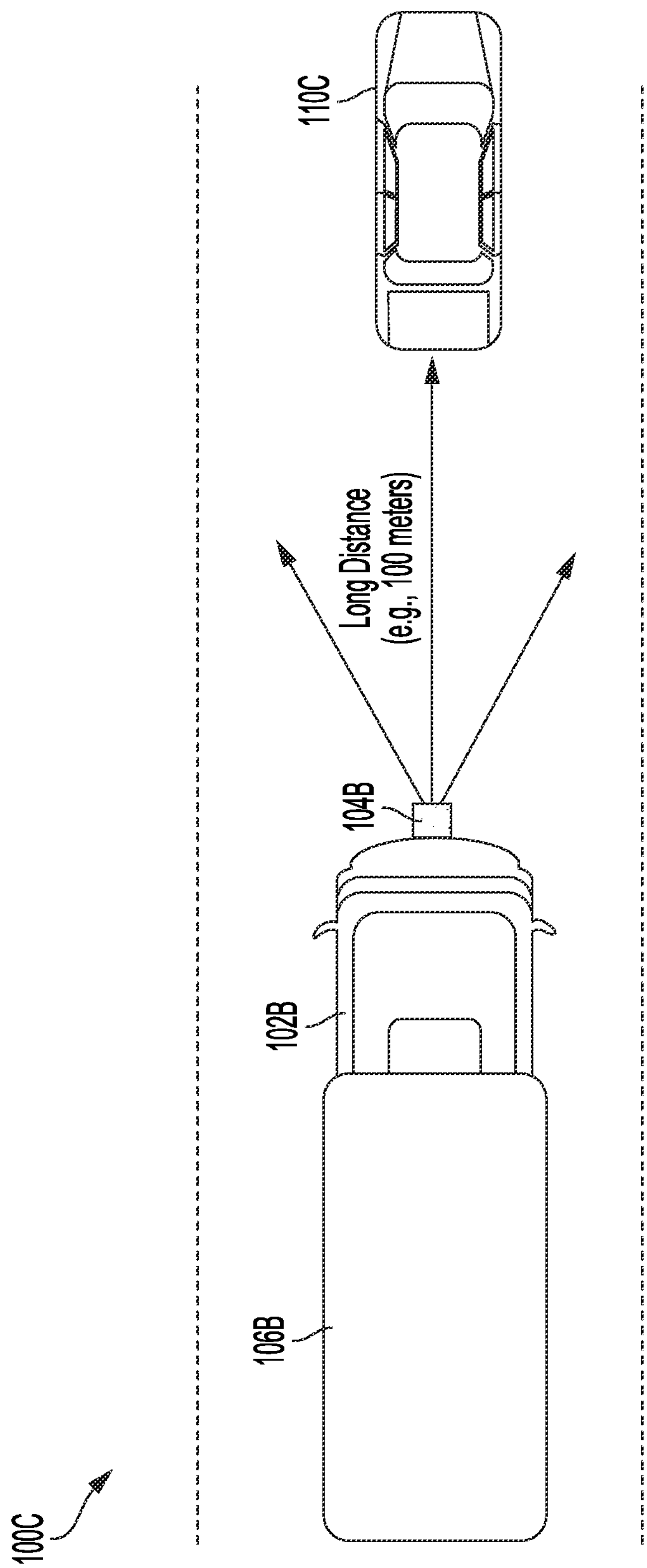


FIG. 3

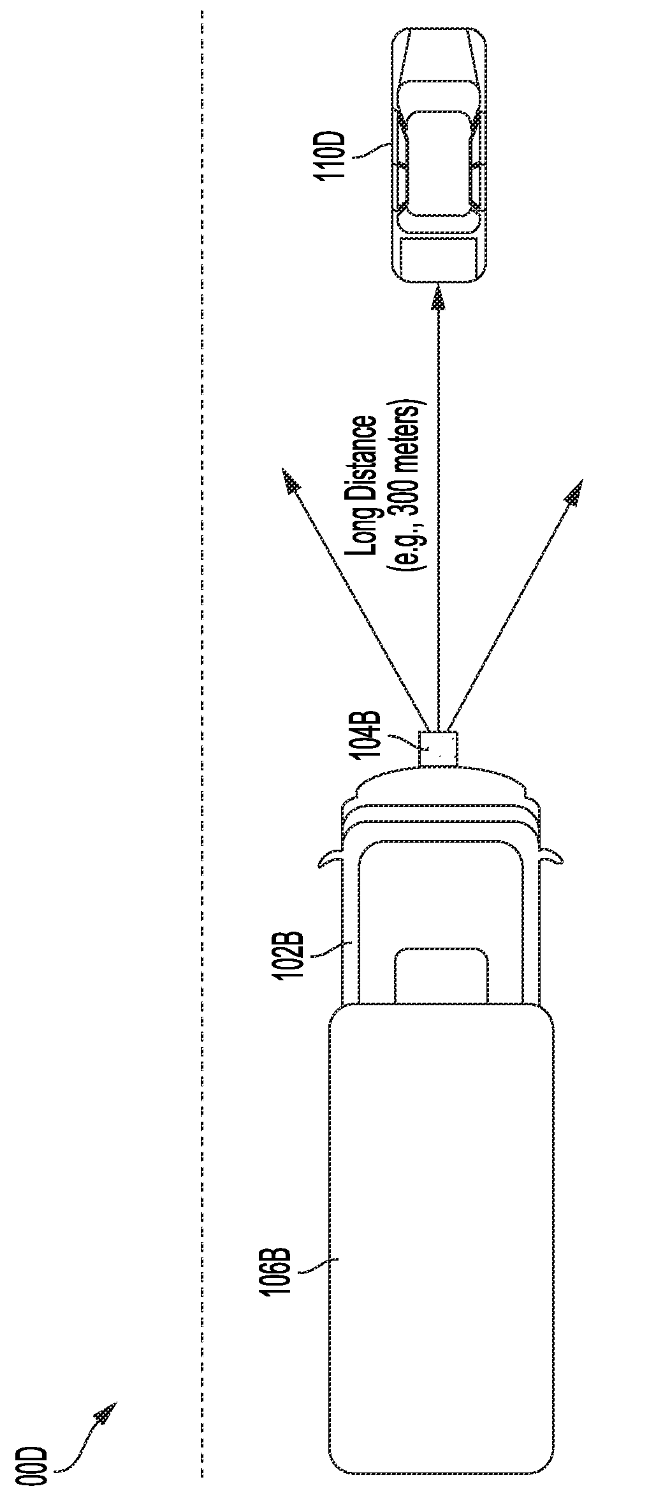
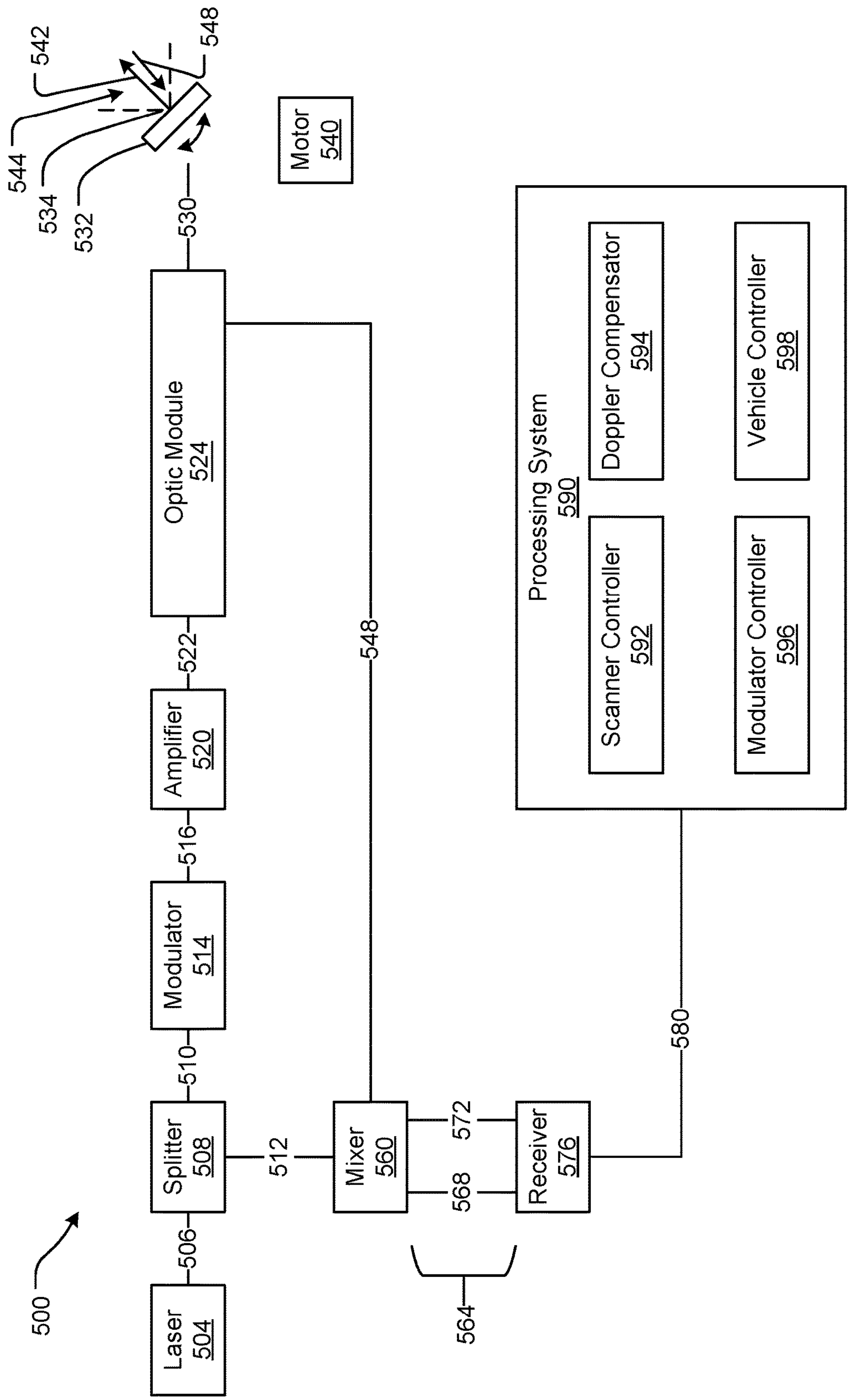


FIG. 4



500
Laser 504
506
Splitter 508
510
Modulator 514
516
Amplifier 520
522
Optic Module 524
530
532
534
544
542
548
Motor 540
548
512
Mixer 560
572
568
Receiver 576
564
580
Processing System 590
Scanner Controller 592
Doppler Compensator 594
Modulator Controller 596
Vehicle Controller 598

FIG. 5

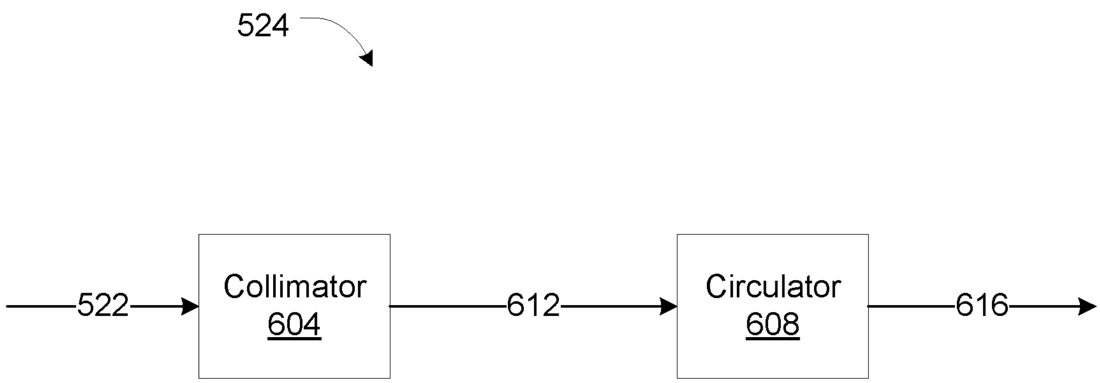


FIG. 6

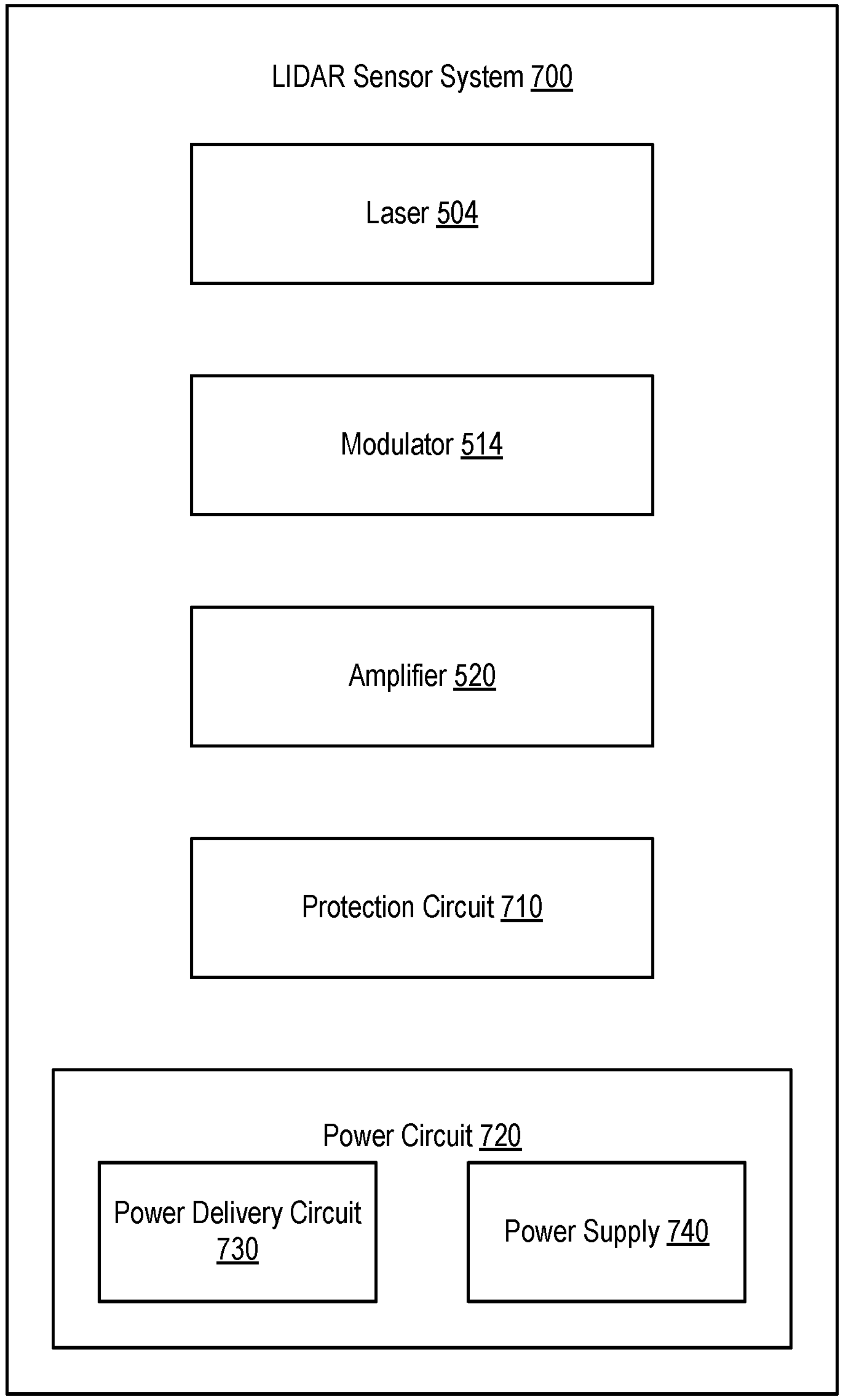
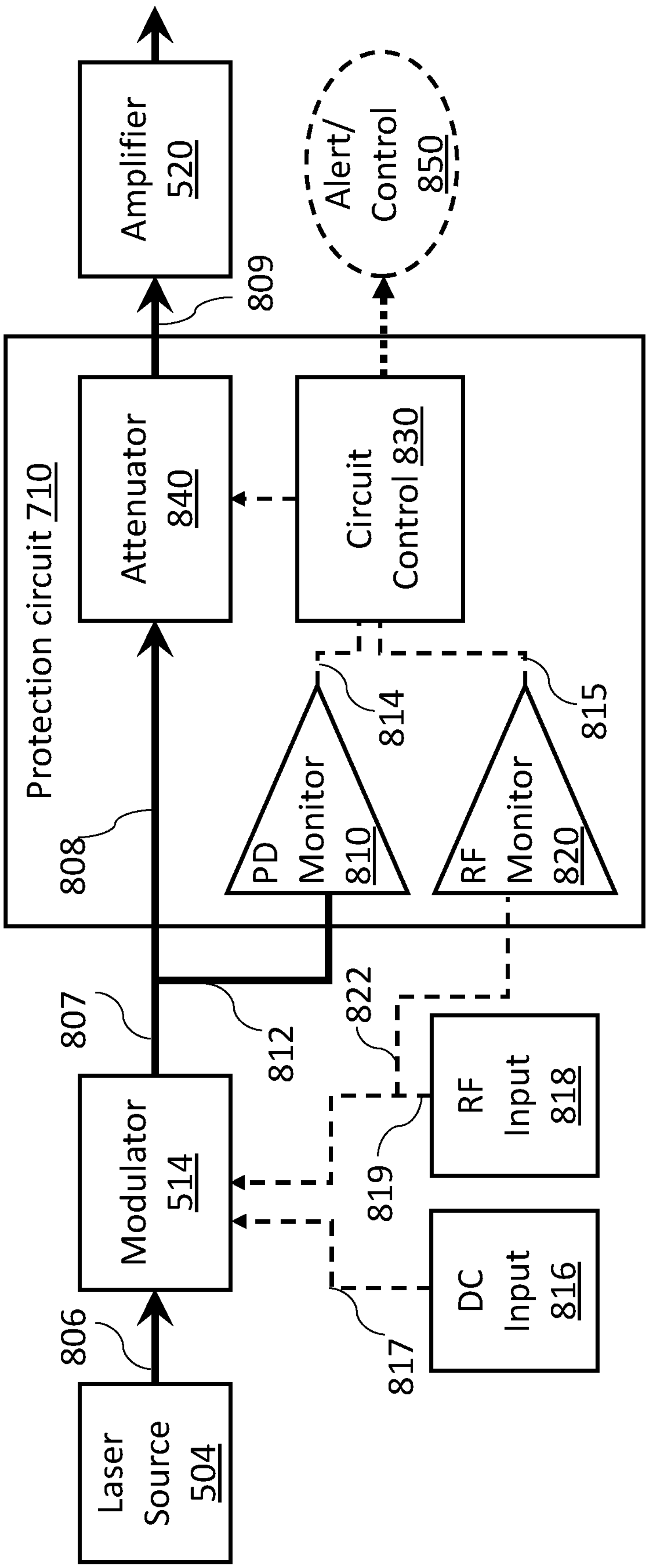

LIDAR Sensor System 700
Laser 504
Modulator 514
Amplifier 520
Protection Circuit 710
Power Circuit 720
Power Delivery Circuit 730
Power Supply 740

FIG. 7


Laser Source 504
806
Modulator 514
807
812
DC Input 816
817
RF Input 818
819
822
Protection circuit 710
808
PD Monitor 810
814
RF Monitor 820
815
Attenuator 840
Circuit Control 830
809
Amplifier 520
Alert/ Control 850

800

FIG. 8

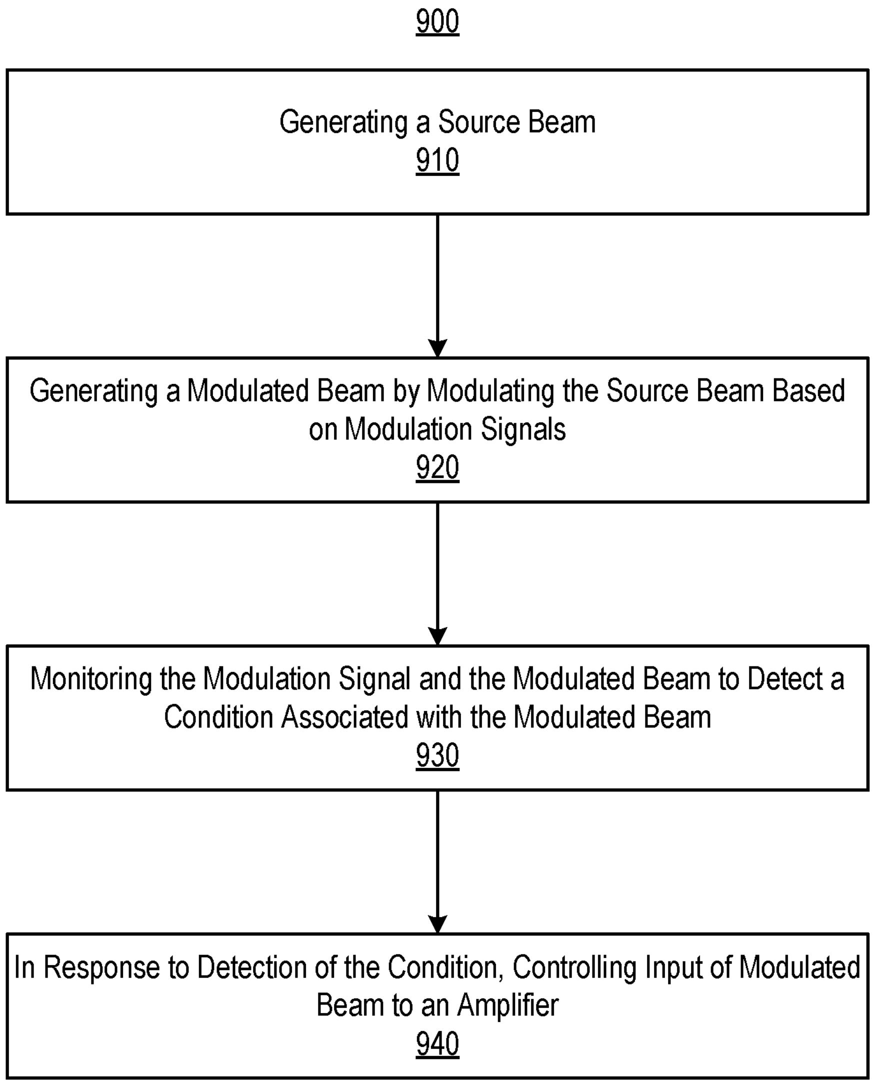

900
Generating a Source Beam
910
Generating a Modulated Beam by Modulating the Source Beam Based on Modulation Signals
920
Monitoring the Modulation Signal and the Modulated Beam to Detect a Condition Associated with the Modulated Beam
930
In Response to Detection of the Condition, Controlling Input of Modulated Beam to an Amplifier
940

FIG. 9

SYSTEMS AND METHODS OF LIDAR SENSOR SYSTEMS HAVING AMPLIFIER PROTECTION CIRCUITS

BACKGROUND

[0001] Optical detection of range using lasers, often referenced by a mnemonic, LIDAR (for "light detection and ranging"), also sometimes referred to as "laser RADAR," is used for a variety of applications, including imaging and collision avoidance. LIDAR provides finer scale range resolution with smaller beam sizes than conventional microwave ranging systems, such as radio-wave detection and ranging (RADAR).

SUMMARY

[0002] At least one aspect relates to a light detection and ranging (LIDAR) sensor system for a vehicle. The LIDAR sensor system includes a laser source, an amplifier, a circuit coupled with the amplifier, and one or more scanning optics. The laser source is configured to output a beam (e.g., laser light). The amplifier is configured to amplify the beam to provide an amplified beam. The circuit is configured to provide power to the amplifier to cause the amplifier to amplify the beam. The circuit is configured to control a parameter of the power. The one or more scanning optics are configured to receive the amplified beam from the amplifier and output the amplified beam.

[0003] At least one aspect relates to an autonomous vehicle control system. The autonomous vehicle control system includes a laser source, an amplifier, a circuit coupled with the amplifier, one or more scanning optics, and one or more processors. The laser source is configured to output a beam. The amplifier is configured to amplify the beam to provide an amplified beam. The circuit is configured to provide power to the amplifier to cause the amplifier to amplify the beam. The circuit is configured to control a parameter of the power. The one or more scanning optics are configured to receive the amplified beam from the amplifier and output the amplified beam. The one or more processors are configured to determine at least one of a range to an object or a velocity of the object based on a return signal from reflection of the amplified beam by the object. The one or more processors are configured to control operation of an autonomous vehicle based on the at least one of the range or the velocity.

[0004] At least one aspect relates to an autonomous vehicle. The autonomous vehicle includes a LIDAR sensor system, a steering system, a braking system, and a vehicle controller. The LIDAR sensor system includes a laser source, an amplifier, a circuit coupled with the amplifier, one or more scanning optics, and one or more processors. The laser source is configured to output a beam. The amplifier is configured to amplify the beam to provide an amplified beam. The circuit is configured to provide power to the amplifier to cause the amplifier to amplify the beam. The circuit is configured to control a parameter of the power. The one or more scanning optics are configured to receive the amplified beam from the amplifier and output the amplified beam. The one or more processors are configured to determine at least one of a range to an object or a velocity of the object based on a return signal from reflection of the amplified beam by the object. The vehicle controller is configured to control operation of at least one of the steering system or the braking system based on the at least one of the range or the velocity.

[0005] Those skilled in the art will appreciate that the summary is illustrative only and is not intended to be in any way limiting. Any of the features described herein may be used with any other features, and any subset of such features can be used in combination according to various embodiments. Other aspects, inventive features, and advantages of the devices and/or processes described herein, as defined solely by the claims, will become apparent in the detailed description set forth herein and taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Implementations are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like reference numerals refer to similar elements and in which:

[0007] FIG. **1** is a block diagram illustrating an example of a system environment for autonomous vehicles according to some implementations;

[0008] FIG. **2** is a block diagram illustrating an example of a system environment for autonomous commercial trucking vehicles according to some implementations;

[0009] FIG. **3** is a block diagram illustrating an example of a system environment for autonomous commercial trucking vehicles according to some implementations;

[0010] FIG. **4** is a block diagram illustrating an example of a system environment for autonomous commercial trucking vehicles according to some implementations;

[0011] FIG. **5** is a block diagram illustrating an example of a LIDAR sensor system according to some implementations;

[0012] FIG. **6** is a block diagram illustrating an example of an optic module of a LIDAR sensor system according to some implementations;

[0013] FIG. **7** is a block diagram illustrating an example of a LIDAR sensor system including a protection circuit according to some implementations;

[0014] FIG. **8** is a block diagram illustrating an example of a LIDAR sensor system including a protection circuit according to some implementations; and

[0015] FIG. **9** is a flow diagram showing an example method for operating a LIDAR sensor system including a protection circuit according to some implementations.

DETAILED DESCRIPTION

[0016] A LIDAR sensor system can generate and transmit a light beam (e.g., laser light) that an object can reflect or otherwise scatter as a return beam corresponding to the transmitted beam. The LIDAR sensor system can receive the return beam, and process the return beam or characteristics thereof to determine parameters regarding the object such as range and velocity. The LIDAR sensor system can apply various amplitude, frequency, and/or or phase modulations to the transmitted beam, which can facilitate relating the return beam to the transmitted beam in order to determine the parameters regarding the object.

[0017] Amplifiers, such as Erbium Doped Fiber Amplifiers (EDFAs) can be used in vehicle LIDAR sensor systems to amplify a light signal. However, various amplifiers including EDFAs can be susceptible to technical deficiencies at high optical power, such as Stimulated Brillouin Scattering (SBS) at high optical power levels (or fluence). For example, when a pulse is missing (e.g., an issue with an electro-optic modulator, etc.), SBS could be initiated. When SBS is present, a system can be more susceptible to damages, particularly in fiber components (e.g., fiber, connectors, EDFA, isolators, etc.).

[0018] Systems and methods in accordance with the present disclosure can provide optical and/or electrical protection of the system. For example, the LIDAR sensor systems can include a protection circuit to detect a condition associated with a modulated beam, and can control input of the modulated beam to an amplifier (e.g., EDFA) in response to a detection of the condition. The condition can include, for example, a missing signal associated with the modulated beam, values of one or more parameters that indicate that operation of the LIDAR system has deviated from a target operation of the LIDAR system that could affect (e.g., damage) the amplifier, or various combinations thereof. For example, the protection circuit can evaluate at least one of the modulation signal or a parameter of the modulated beam. In response to a detection of the condition, the protection circuit can control an optical attenuator to eliminate the input to the amplifier (e.g., EDFA). By controlling the input to the amplifier in response to a detection of such a condition, damages to the system (e.g., optical components) can be prevented, thereby allowing for reliable operation of the system.

[0019] In some implementations, the LIDAR sensor system includes a power circuit, such as a power control circuit, to control delivery of power to the EDFA and/or to control the input light (e.g., beam or modulated beam) to the EDFA. For example, the protection circuit and/or the power circuit can be used to facilitate at least one of control of input light to the EDFA, turning off the EDFA during use of the protection circuit, or shaping of the output light to shape the output optical pulse of the EDFA. For example, the LIDAR sensor system, responsive to and/or immediately after detection of the condition (e.g., missing pulse), can control the input light to the EDFA (e.g., cause attenuation for the input light, such as to cause a reverse bias to one or more semiconductor optical amplifiers (SOAs) associated with the input light) and/or turn off the EDFA for a period (e.g., about 100 milliseconds or more). This can be useful, for example, where the LIDAR sensor system is implemented as a quasi-continuous wave (QCW) system. The power circuit can more effectively trigger operation of the EDFA, such as to turn the EDFA on/off, based on a waveform of the current delivered to the EDFA by the power circuit. For example, as compared to systems that provide a large instantaneous pulse at a front end of the current to the EDFA, followed by exponential decay, systems and methods in accordance with the present disclosure can have a more smooth input light delivery to the EDFA, such as facilitate EDFA amplification being performed more slowly. This can avoid or reduce the likelihood of SBS for operation of the EDFA, such as in relation to the shape of the output optical pulse of the EDFA.

System Environments for Autonomous Vehicles

[0020] FIG. **1** is a block diagram illustrating an example of a system environment for autonomous vehicles according to some implementations. FIG. **1** depicts an example autonomous vehicle **100** within which the various techniques disclosed herein may be implemented. The vehicle **100**, for example, may include a powertrain **102** including a prime mover **104** powered by an energy source **106** and capable of providing power to a drivetrain **108**, as well as a control system **110** including a direction control **112**, a powertrain control **114**, and a brake control **116**. The vehicle **100** may be implemented as any number of different types of vehicles, including vehicles capable of transporting people and/or cargo, and capable of traveling in various environments. The aforementioned components **102-116** can vary widely based upon the type of vehicle within which these components are utilized, such as a wheeled land vehicle such as a car, van, truck, or bus. The prime mover **104** may include one or more electric motors and/or an internal combustion engine (among others). The energy source may include, for example, a fuel system (e.g., providing gasoline, diesel, hydrogen, etc.), a battery system, solar panels or other renewable energy source, and/or a fuel cell system. The drivetrain **108** can include wheels and/or tires along with a transmission and/or any other mechanical drive components to convert the output of the prime mover **104** into vehicular motion, as well as one or more brakes configured to controllably stop or slow the vehicle **100** and direction or steering components suitable for controlling the trajectory of the vehicle **100** (e.g., a rack and pinion steering linkage enabling one or more wheels of the vehicle **100** to pivot about a generally vertical axis to vary an angle of the rotational planes of the wheels relative to the longitudinal axis of the vehicle). In some implementations, combinations of powertrains and energy sources may be used (e.g., in the case of electric/gas hybrid vehicles), and in some instances multiple electric motors (e.g., dedicated to individual wheels or axles) may be used as a prime mover.

[0021] The direction control **112** may include one or more actuators and/or sensors for controlling and receiving feedback from the direction or steering components to enable the vehicle **100** to follow a desired trajectory. The powertrain control **114** may be configured to control the output of the powertrain **102**, e.g., to control the output power of the prime mover **104**, to control a gear of a transmission in the drivetrain **108**, etc., thereby controlling a speed and/or direction of the vehicle **100**. The brake control **116** may be configured to control one or more brakes that slow or stop vehicle **100**, e.g., disk or drum brakes coupled to the wheels of the vehicle.

[0022] Other vehicle types, including but not limited to off-road vehicles, all-terrain or tracked vehicles, construction equipment, may utilize different powertrains, drivetrains, energy sources, direction controls, powertrain controls and brake controls. Moreover, in some implementations, some of the components can be combined, e.g., where directional control of a vehicle is primarily handled by varying an output of one or more prime movers.

[0023] Various levels of autonomous control over the vehicle **100** can be implemented in a vehicle control system **120**, which may include one or more processors **122** and one or more memories **124**, with each processor **122** configured to execute program code instructions **126** stored in a memory **124**. The processor(s) can include, for example, graphics processing unit(s) ("GPU(s)")) and/or central processing unit(s) ("CPU(s)").

[0024] Sensors **130** may include various sensors suitable for collecting information from a vehicle's surrounding environment for use in controlling the operation of the vehicle. For example, sensors **130** can include radar sensor **134**, LIDAR (Light Detection and Ranging) sensor **136**, a

3D positioning sensors **138**, e.g., any of an accelerometer, a gyroscope, a magnetometer, or a satellite navigation system such as GPS (Global Positioning System), GLONASS (Globalnaya Navigazionnaya Sputnikovaya Sistema, or Global Navigation Satellite System), BeiDou Navigation Satellite System (BDS), Galileo, Compass, etc. The 3D positioning sensors **138** can be used to determine the location of the vehicle on the Earth using satellite signals. The sensors **130** can include a camera **140** and/or an IMU (inertial measurement unit) **142**. The camera **140** can be a monographic or stereographic camera and can record still and/or video images. The IMU **142** can include multiple gyroscopes and accelerometers capable of detecting linear and rotational motion of the vehicle in three directions. One or more encoders (not illustrated), such as wheel encoders may be used to monitor the rotation of one or more wheels of vehicle **100**. Each sensor **130** can output sensor data at various data rates, which may be different than the data rates of other sensors **130**.

[0025] The outputs of sensors **130** may be provided to a set of control subsystems **150**, including a localization subsystem **152**, a planning subsystem **156**, a perception subsystem **154**, and a control subsystem **158**. The localization subsystem **152** can perform functions such as precisely determining the location and orientation (also sometimes referred to as "pose") of the vehicle **100** within its surrounding environment, and generally within some frame of reference. The location of an autonomous vehicle can be compared with the location of an additional vehicle in the same environment as part of generating labeled autonomous vehicle data. The perception subsystem **154** can perform functions such as detecting, tracking, determining, and/or identifying objects within the environment surrounding vehicle **100**. A machine learning model in accordance with some implementations can be utilized in tracking objects. The planning subsystem **156** can perform functions such as planning a trajectory for vehicle **100** over some timeframe given a desired destination as well as the static and moving objects within the environment. A machine learning model in accordance with some implementations can be utilized in planning a vehicle trajectory. The control subsystem **158** can perform functions such as generating suitable control signals for controlling the various controls in the vehicle control system **120** in order to implement the planned trajectory of the vehicle **100**. A machine learning model can be utilized to generate one or more signals to control an autonomous vehicle to implement the planned trajectory.

[0026] Multiple sensors of types illustrated in FIG. **1** can be used for redundancy and/or to cover different regions around a vehicle, and other types of sensors may be used. Various types and/or combinations of control subsystems may be used. Some or all of the functionality of a subsystem **152-158** may be implemented with program code instructions **126** resident in one or more memories **124** and executed by one or more processors **122**, and these subsystems **152-158** may in some instances be implemented using the same processor(s) and/or memory. Subsystems may be implemented at least in part using various dedicated circuit logic, various processors, various field programmable gate arrays ("FPGA"), various application-specific integrated circuits ("ASIC"), various real time controllers, and the like, as noted above, multiple subsystems may utilize circuitry, processors, sensors, and/or other components. Further, the various components in the vehicle control system **120** may be networked in various manners.

[0027] In some implementations, the vehicle **100** may also include a secondary vehicle control system (not illustrated), which may be used as a redundant or backup control system for the vehicle **100**. In some implementations, the secondary vehicle control system may be capable of fully operating the autonomous vehicle **100** in the event of an adverse event in the vehicle control system **120**, while in other implementations, the secondary vehicle control system may only have limited functionality, e.g., to perform a controlled stop of the vehicle **100** in response to an adverse event detected in the primary vehicle control system **120**. In still other implementations, the secondary vehicle control system may be omitted.

[0028] Various architectures, including various combinations of software, hardware, circuit logic, sensors, and networks, may be used to implement the various components illustrated in FIG. **1**. Each processor may be implemented, for example, as a microprocessor and each memory may represent the random access memory ("RAM") devices comprising a main storage, as well as any supplemental levels of memory, e.g., cache memories, non-volatile or backup memories (e.g., programmable or flash memories), read-only memories, etc. In addition, each memory may be considered to include memory storage physically located elsewhere in the vehicle **100**, e.g., any cache memory in a processor, as well as any storage capacity used as a virtual memory, e.g., as stored on a mass storage device or another computer controller. One or more processors illustrated in FIG. **1**, or entirely separate processors, may be used to implement additional functionality in the vehicle **100** outside of the purposes of autonomous control, e.g., to control entertainment systems, to operate doors, lights, convenience features, etc.

[0029] In addition, for additional storage, the vehicle **100** may include one or more mass storage devices, e.g., a removable disk drive, a hard disk drive, a direct access storage device ("DASD"), an optical drive (e.g., a CD drive, a DVD drive, etc.), a solid state storage drive ("SSD"), network attached storage, a storage area network, and/or a tape drive, among others.

[0030] Furthermore, the vehicle **100** may include a user interface **164** to enable vehicle **100** to receive a number of inputs from and generate outputs for a user or operator, e.g., one or more displays, touchscreens, voice and/or gesture interfaces, buttons and other tactile controls, etc. Otherwise, user input may be received via another computer or electronic device, e.g., via an app on a mobile device or via a web interface.

[0031] Moreover, the vehicle **100** may include one or more network interfaces, e.g., network interface **162**, suitable for communicating with one or more networks **170** (e.g., a Local Area Network ("LAN"), a wide area network ("WAN"), a wireless network, and/or the Internet, among others) to permit the communication of information with other computers and electronic device, including, for example, a central service, such as a cloud service, from which the vehicle **100** receives environmental and other data for use in autonomous control thereof. Data collected by the one or more sensors **130** can be uploaded to a computing system **172** via the network **170** for additional processing. In some implementations, a time stamp can be added to each instance of vehicle data prior to uploading.

[0032] Each processor illustrated in FIG. **1**, as well as various additional controllers and subsystems disclosed herein, generally operates under the control of an operating system and executes or otherwise relies upon various computer software applications, components, programs, objects, modules, data structures, etc., as will be described in greater detail below. Moreover, various applications, components, programs, objects, modules, etc. may also execute on one or more processors in another computer coupled to vehicle **100** via network **170**, e.g., in a distributed, cloud-based, or client-server computing environment, whereby the processing required to implement the functions of a computer program may be allocated to multiple computers and/or services over a network.

[0033] In general, the routines executed to implement the various implementations described herein, whether implemented as part of an operating system or a specific application, component, program, object, module or sequence of instructions, or even a subset thereof, will be referred to herein as "program code". Program code can include one or more instructions that are resident at various times in various memory and storage devices, and that, when read and executed by one or more processors, perform the steps necessary to execute steps or elements embodying the various aspects of the present disclosure. Moreover, while implementations have and hereinafter will be described in the context of fully functioning computers and systems, it will be appreciated that the various implementations described herein are capable of being distributed as a program product in a variety of forms, and that implementations can be implemented regardless of the particular type of computer readable media used to actually carry out the distribution.

[0034] Examples of computer readable media include tangible, non-transitory media such as volatile and non-volatile memory devices, floppy and other removable disks, solid state drives, hard disk drives, magnetic tape, and optical disks (e.g., CD-ROMs, DVDs, etc.) among others.

[0035] In addition, various program code described hereinafter may be identified based upon the application within which it is implemented in a specific implementation. Any particular program nomenclature that follows is used merely for convenience, and thus the present disclosure should not be limited to use solely in any specific application identified and/or implied by such nomenclature. Furthermore, given the typically endless number of manners in which computer programs may be organized into routines, procedures, methods, modules, objects, and the like, as well as the various manners in which program functionality may be allocated among various software layers that are resident within a typical computer (e.g., operating systems, libraries, API's, applications, applets, etc.), the present disclosure is not limited to the specific organization and allocation of program functionality described herein.

1. LIDAR for Automotive Applications

[0036] A truck can include a LIDAR system (e.g., vehicle control system **120** in FIG. **1**, LIDAR sensor system **500** in FIG. **5**, among others described herein). In some implementations, the LIDAR sensor system **500** can use frequency modulation to encode an optical signal and scatter the encoded optical signal into free-space using optics. By detecting the frequency differences between the encoded optical signal and a returned signal reflected back from an object, the frequency modulated (FM) LIDAR sensor system can determine the location of the object and/or precisely measure the velocity of the object using the Doppler effect. In some implementations, an FM LIDAR sensor system may use a continuous wave (referred to as, "FMCW LIDAR") or a quasi-continuous wave (referred to as, "FMQCW LIDAR"). In some implementations, the LIDAR sensor system can use amplitude modulation (AM) and/or phase modulation (PM) to encode an optical signal and scatters the encoded optical signal into free-space using optics.

[0037] In some instances, an object (e.g., a pedestrian wearing dark clothing) may have a low reflectivity, in that it only reflects back to the sensors (e.g., sensors **130** in FIG. **1**) of the FM or PM LIDAR sensor system a low amount (e.g., 10% or less) of the light that hit the object. In other instances, an object (e.g., a shiny road sign) may have a high reflectivity (e.g., above 10%), in that it reflects back to the sensors of the FM LIDAR sensor system a high amount of the light that hit the object.

[0038] Regardless of the object's reflectivity, an FM LIDAR sensor system may be able to detect (e.g., classify, recognize, discover, etc.) the object at greater distances (e.g., 2×) than a conventional LIDAR sensor system. For example, an FM LIDAR sensor system may detect a low reflectivity object beyond 300 meters, and a high reflectivity object beyond 400 meters.

[0039] To achieve such improvements in detection capability, the FM LIDAR sensor system may use sensors (e.g., sensors **130** in FIG. **1**). In some implementations, these sensors can be single photon sensitive, meaning that they can detect the smallest amount of light possible. While an FM LIDAR sensor system may, in some applications, use infrared wavelengths (e.g., 950 nm, 1550 nm, etc.), it is not limited to the infrared wavelength range (e.g., near infrared: 800 nm-1500 nm; middle infrared: 1500 nm-5600 nm; and far infrared: 5600 nm-1,000,000 nm). By operating the FM or PM LIDAR sensor system in infrared wavelengths, the FM or PM LIDAR sensor system can broadcast stronger light pulses or light beams than conventional LIDAR sensor systems.

[0040] Thus, by detecting an object at greater distances, an FM LIDAR sensor system may have more time to react to unexpected obstacles. Indeed, even a few milliseconds of extra time could improve response time and comfort, especially with heavy vehicles (e.g., commercial trucking vehicles) that are driving at highway speeds.

[0041] The FM LIDAR sensor system can provide accurate velocity for each data point instantaneously. In some implementations, a velocity measurement is accomplished using the Doppler effect which shifts frequency of the light received from the object based at least one of the velocity in the radial direction (e.g., the direction vector between the object detected and the sensor) or the frequency of the laser signal. For example, for velocities encountered in on-road situations where the velocity is less than 100 meters per second (m/s), this shift at a wavelength of 1550 nanometers (nm) amounts to the frequency shift that is less than 130 megahertz (MHz). This frequency shift is small such that it is difficult to detect directly in the optical domain. However, by using coherent detection in FMCW, PMCW, or FMQCW LIDAR sensor systems, the signal can be converted to the RF domain such that the frequency shift can be calculated using various signal processing techniques. This enables the autonomous vehicle control system to process incoming data faster.

[0042] Instantaneous velocity calculation also makes it easier for the FM LIDAR sensor system to determine distant or sparse data points as objects and/or track how those objects are moving over time. For example, an FM LIDAR sensor (e.g., sensors **130** in FIG. **1**) may only receive a few returns (e.g., hits) on an object that is 300 m away, but if those return give a velocity value of interest (e.g., moving towards the vehicle at >70 mph), then the FM LIDAR sensor system and/or the autonomous vehicle control system may determine respective weights to probabilities associated with the objects.

[0043] Faster identification and/or tracking of the FM LIDAR sensor system gives an autonomous vehicle control system more time to maneuver a vehicle. A better understanding of how fast objects are moving also allows the autonomous vehicle control system to plan a better reaction.

[0044] The FM LIDAR sensor system can have less static compared to conventional LIDAR sensor systems. That is, the conventional LIDAR sensor systems that are designed to be more light-sensitive typically perform poorly in bright sunlight. These systems also tend to suffer from crosstalk (e.g., when sensors get confused by each other's light pulses or light beams) and from self-interference (e.g., when a sensor gets confused by its own previous light pulse or light beam). To overcome these disadvantages, vehicles using the conventional LIDAR sensor systems often need extra hardware, complex software, and/or more computational power to manage this "noise."

[0045] In contrast, FM LIDAR sensor systems do not suffer from these types of issues because each sensor is specially designed to respond only to its own light characteristics (e.g., light beams, light waves, light pulses). If the returning light does not match the timing, frequency, and/or wavelength of what was originally transmitted, then the FM sensor can filter (e.g., remove, ignore, etc.) out that data point. As such, FM LIDAR sensor systems produce (e.g., generates, derives, etc.) more accurate data with less hardware or software requirements, enabling smoother driving.

[0046] The FM LIDAR sensor system can be easier to scale than conventional LIDAR sensor systems. As more self-driving vehicles (e.g., cars, commercial trucks, etc.) show up on the road, those powered by an FM LIDAR sensor system likely will not have to contend with interference issues from sensor crosstalk. Furthermore, an FM LIDAR sensor system uses less optical peak power than conventional LIDAR sensors. As such, some or all of the optical components for an FM LIDAR can be produced on a single chip, which produces its own benefits, as discussed herein.

[0047] FIG. **2** is a block diagram illustrating an example of a system environment for autonomous commercial trucking vehicles, according to some implementations. The environment **100**B includes a commercial truck **102**B for hauling cargo **106**B. In some implementations, the commercial truck **102**B may include vehicles configured to long-haul freight transport, regional freight transport, intermodal freight transport (i.e., in which a road-based vehicle is used as one of multiple modes of transportation to move freight), and/or any other road-based freight transport applications. In some implementations, the commercial truck **102**B may be a flatbed truck, a refrigerated truck (e.g., a reefer truck), a vented van (e.g., dry van), a moving truck, etc. In some implementations, the cargo **106**B may be goods and/or products. In some implementations, the commercial truck **102**B may include a trailer to carry the cargo **106**B, such as a flatbed trailer, a lowboy trailer, a step deck trailer, an extendable flatbed trailer, a sidekit trailer, etc.

[0048] The environment **100**B includes an object **110**B (shown in FIG. **2** as another vehicle) that is within a distance range that is equal to or less than 30 meters from the truck.

[0049] The commercial truck **102**B may include a LIDAR sensor system **104**B (e.g., an FM LIDAR sensor system, vehicle control system **120** in FIG. **1**, LIDAR sensor system **500** in FIG. **5**) for determining a distance to the object **110**B and/or measuring the velocity of the object **110**B. Although FIG. **2** shows that one LIDAR sensor system **104**B is mounted on the front of the commercial truck **102**B, the number of LIDAR sensor systems and the mounting area of the LIDAR sensor system on the commercial truck are not limited to a particular number or a particular area. The commercial truck **102**B may include any number of LIDAR sensor systems **104**B (or components thereof, such as sensors, modulators, coherent signal generators, etc.) that are mounted onto any area (e.g., front, back, side, top, bottom, underneath, and/or bottom) of the commercial truck **102**B to facilitate the detection of an object in any free-space relative to the commercial truck **102**B.

[0050] As shown, the LIDAR sensor system **104**B in environment **100**B may be configured to detect an object (e.g., another vehicle, a bicycle, a tree, street signs, potholes, etc.) at short distances (e.g., 30 meters or less) from the commercial truck **102**B.

[0051] FIG. **3** is a block diagram illustrating an example of a system environment for autonomous commercial trucking vehicles, according to some implementations. The environment **100**C includes the same components (e.g., commercial truck **102**B, cargo **106**B, LIDAR sensor system **104**B, etc.) that are included in environment **100**B.

[0052] The environment **100**C includes an object **110**C (shown in FIG. **3** as another vehicle) that is within a distance range that is (i) more than 30 meters and (ii) equal to or less than 150 meters from the commercial truck **102**B. As shown, the LIDAR sensor system **104**B in environment **100**C may be configured to detect an object (e.g., another vehicle, a bicycle, a tree, street signs, potholes, etc.) at a distance (e.g., 100 meters) from the commercial truck **102**B.

[0053] FIG. **4** is a block diagram illustrating an example of a system environment for autonomous commercial trucking vehicles, according to some implementations. The environment **100**D includes the same components (e.g., commercial truck **102**B, cargo **106**B, LIDAR sensor system **104**B, etc.) that are included in environment **100**B.

[0054] The environment **100**D includes an object **110**D (shown in FIG. **4** as another vehicle) that is within a distance range that is more than 150 meters from the commercial truck **102**B. As shown, the LIDAR sensor system **104**B in environment **100**D may be configured to detect an object (e.g., another vehicle, a bicycle, a tree, street signs, potholes, etc.) at a distance (e.g., 300 meters) from the commercial truck **102**B.

[0055] In commercial trucking applications, it is important to effectively detect objects at all ranges due to the increased weight and, accordingly, longer stopping distance required for such vehicles. FM LIDAR sensor systems (e.g., FMCW and/or FMQW systems) or PM LIDAR sensor systems are well-suited for commercial trucking applications due to the advantages described above. As a result, commercial trucks equipped with such systems may have an enhanced ability to move both people and goods across short or long distances. In various implementations, such FM or PM LIDAR sensor systems can be used in semi-autonomous applications, in which the commercial truck has a driver and some functions of the commercial truck are autonomously operated using the FM or PM LIDAR sensor system, or fully autonomous applications, in which the commercial truck is operated entirely by the FM or LIDAR sensor system, alone or in combination with other vehicle systems.

2. LIDAR Sensor Systems

[0056] FIG. **5** is a block diagram illustrating an example of a LIDAR sensor system **500** according to some implementations. The LIDAR sensor system **500** can be used to determine parameters regarding objects, such as range and velocity, and output the parameters to a remote system. For example, the LIDAR sensor system **500** can output the parameters for use by a vehicle controller that can control operation of a vehicle responsive to the received parameters (e.g., vehicle controller **598**) or a display that can present a representation of the parameters. The LIDAR sensor system **500** can be a coherent detection system. The LIDAR sensor system **500** can be used to implement various features and components of the systems described with reference to FIGS. **1-4**. The LIDAR sensor system **500** can include components for performing various detection approaches, such as to be operated as an amplitude modular LIDAR system or a coherent LIDAR system. The LIDAR sensor system **500** can be used to perform time of flight range determination. In some implementations, various components or combinations of components of the LIDAR sensor system **500**, such as laser source **504** and modulator **514**, can be in the same housing, provided in the same circuit board or other electronic component, or otherwise integrated. In some implementations, various components or combinations of components of the LIDAR sensor system **500** can be provided as separate components, such as by using optical couplings (e.g., optical fibers) for components that generate and/or receive optical signals, such as light beams, or wired or wireless electronic connections for components that generate and/or receive electrical (e.g., data) signals. Various components of the LIDAR sensor system **500** can be arranged with respect to one another such that light (e.g., beams of light) between the components is directed through free space, such as a space provided by an air (or vacuum) gap, a space that is not through an optical fiber, a space that is free of structural components around a path along which the light is directed (e.g., an empty space at least on the order of millimeters away from a direct line path between the components; an empty space of a size greater than an expected beam width of the light, such as where the light is a collimated beam), or various combinations thereof.

[0057] In some implementations, a semiconductor substrate and/or semiconductor package include one or more components of at least one of a transmission (Tx) path or a receiving (Rx) path of the LIDAR sensor system **500**. This can include, for example, optical and/or electronic components that can generate heat that may be transferred into the semiconductor substrate and/or semiconductor package during operation. In some implementations, the semiconductor substrate and/or semiconductor package include at least one of silicon photonics circuitry, planar lightwave circuitry (PLC), or III-V semiconductor circuitry.

[0058] In some implementations, the optical and/or electronic components formed on or coupled to the semiconductor substrate and/or semiconductor package to perform a plurality of functions in the LIDAR sensor system **500** are collectively referred to as a circuit module. In some implementations, the circuit module includes III-V semiconductor circuitry coupled to at least one of silicon photonics circuitry or PLC. In the present disclosure, "coupling" may refer to a physical connection, an electrical connection, or both, between two components.

[0059] In some implementations, a first semiconductor substrate and/or a first semiconductor package include the Tx path and a second semiconductor substrate and/or a second semiconductor package may include the Rx path. In some arrangements, the Rx input/output ports and/or the Tx input/output ports may occur (or be formed/disposed/located/placed) along one or more edges of one or more semiconductor substrates and/or semiconductor packages.

[0060] In some implementations, the circuit module include at least one of silicon photonics circuitry, PLC, or III-V semiconductor circuitry in which all of its components (e.g., optical paths, optical amplifiers, phase modulators, etc.) are formed on, disposed over, or otherwise coupled to a single substrate. In some implementations, all of the components of the circuit module are formed on, disposed over, or otherwise coupled to a single layer to form a horizontal structure of an integrated circuit. In some implementations, components of the circuit module are formed on, disposed over, or otherwise coupled to multiple layers stacked on a single substrate to form a vertical structure of an integrated circuit. For example, the circuit module may include phase or amplitude modulators implemented as one or more PLC modules, optical paths implemented as silicon photonics circuitry, and SOAs implemented as III-V modules, all of which are formed on, disposed over, or otherwise coupled to a single III-V substrate. The III-V semiconductor materials may include at least one of gallium arsenide (GaAs), indium phosphide (InP), indium arsenide (InAs), or combinations thereof.

[0061] The LIDAR sensor system **500** can include a laser source **504** that generates and emits a beam **506**, such as a carrier wave light beam. An optic element **508** can split the beam **506** into a beam **510** (sometimes referred to as input beam) and a reference beam **512** (e.g., reference signal). In some implementations, any suitable optical, electronic, or optoelectronic elements are used to provide the beam **510** and the reference beam **512** from the laser source **504** to other elements. For example, the optic element **508** can be a splitter or a circulator.

[0062] A modulator **514** can modulate one or more properties of the input beam **510** to generate a beam **516** (e.g., target beam) and/or encode information on the input beam **510**. In some implementations, the modulator **514** can modulate a frequency of the input beam **510** (e.g., optical frequency corresponding to optical wavelength, where $c=\lambda\nu$, where c is the speed of light, λ is the wavelength, and ν is the frequency). For example, the modulator **514** can modulate a frequency of the input beam **510** linearly such that a frequency of the beam **516** increases or decreases linearly over time. As another example, the modulator **514** can modulate a frequency of the input beam **510** non-linearly (e.g., exponentially). In some implementations, the modulator **514** can modulate a phase of the input beam **510** to generate the beam **516**. However, the modulation techniques are not limited to the frequency modulation and the phase modulation. Any suitable modulation techniques can be used to modulate one or more properties of a beam. Returning to FIG. **5**, the modulator **514** can modulate the beam **510** subsequent to splitting of the beam **506** by the optic element **508**, such that the reference beam **512** is unmodulated, or the modulator **514** can modulate the beam **506** and provide a modulated beam to the optic element **508** for the optic element **508** to split into a target beam and a reference beam. In some implementations, the modulator **514** includes a circuit module having at least one of silicon photonics circuitry, PLC, or III-V semiconductor circuitry. In some implementations, the modulator **514** may be a Mach-Zehnder modulator. In some implementations, the modulator **514** can control one or more properties of the light in continuous wave operation of the LIDAR sensor system **500**. In some implementations, the modulator **514** can control one or more properties of light in quasi continuous wave operation of the LIDAR sensor system **500**.

[0063] The beam **516**, which is used for outputting a transmitted signal, can have most of the energy of the beam **506** outputted by the laser source **504**, while the reference beam **512** can have significantly less energy, yet sufficient energy to enable mixing with a return beam **548** (e.g., returned light) scattered from an object. The reference beam **512** can be used as a local oscillator (LO) signal. The reference beam **512** passes through a reference path and can be provided to a mixer **560**. An amplifier **520** can amplify the beam **516** to output a beam **522**.

[0064] The LIDAR sensor system **500** can include an optic module **524**, which can receive the beam **522**. The optic module **524** can be a free space optic. For example, the optic module **524** can include one or more optics (e.g., lenses, mirrors, waveguides, grating couplers, prisms, waveplates) arranged to have a gap (e.g., air gap) between the one or more optics, allowing for free space transmission of light (e.g., rather than all light being coupled between optics by fibers). The optic module **524** can perform functions such as collimating, filtering, and/or polarizing the beam **522** to output a beam **530** to optics **532** (e.g., scanning optics).

[0065] The LIDAR sensor system **500** can include a protection circuit (not shown) as discussed with respect to FIG. **7** and FIG. **8**. The protection circuit can be operably coupled to the modulator **514** and the amplifier **520**. The protection circuit can detect a condition associated with the beam **516** (e.g., a modulated beam) and control the electrical input to the amplifier **520** based on the detection of the condition.

[0066] Referring to FIG. **6**, the optic module **524** can include at least one collimator **604** and at least one circulator **608**. For example, the circulator **608** can be between the collimator **604** and the optics **532** of FIG. **5**. The circulator **608** can receive a collimated beam **612** outputted by the collimator **604** and output a beam **616** (e.g., the beam **530** depicted in FIG. **5**) to the optics **532**. In some implementations, the circulator **608** can be between the laser source **504** and the collimator **604**. At least one of the collimator **604** or the circulator **608** can be free space optics (and can be coupled with one another in free space), such as by being optically coupled via air gaps rather than optical fibers.

[0067] Referring further to FIG. **5**, the optic module **524** can receive return beam **548** from the optics **532** and provide the return beam **548** to the mixer **560**. The optics **532** can be scanning optics, such as one or more steering mirrors or polygon reflectors or deflectors to adjust the angle of received beams relative to outputted beams based on the orientation of outer surfaces (e.g., facets) of the optics relative to the received beam, or solid-state components (e.g., phased arrays, electro-optic crystals) configured to modify the direction of received light.

[0068] The optics **532** can define a field of view **544** that corresponds to angles scanned (e.g., swept) by the beam **542** (e.g., a transmitted beam). For example, the beam **542** can be scanned in the particular plane, such as an azimuth plane or elevation plane (e.g., relative to an object to which the LIDAR sensor system **500** is coupled, such as an autonomous vehicle). The optics **532** can be oriented so that the field of view **544** sweeps an azimuthal plane relative to the optics **532**.

[0069] At least one motor **540** can be coupled with the optics **532** to control at least one of a position or an orientation of the optics **532** relative to the beam **530**. For example, where the optics **532** include a mirror, reflector, or deflector, the motor **540** can rotate the optics **532** relative to an axis **534** (e.g., an axis orthogonal to the frame of reference depicted in FIG. **5**) so that surfaces of the optics **532** at which the beam **530** is received vary in angle or orientation relative to the beam **530**, causing the beam **542** to be varied in angle or direction as the beam **542** is outputted from the optics **532**.

[0070] The beam **542** can be outputted from the optics **532** and reflected or otherwise scattered by an object (not shown) as a return beam **548** (e.g., return signal). The return beam **548** can be received on a reception path, which can include the circulator **608**, and provided to the mixer **560**.

[0071] The mixer **560** can be an optical hybrid, such as a 90 degree optical hybrid. The mixer **560** can receive the reference beam **512** and the return beam **548**, and mix the reference beam **512** and the return beam **548** to output a signal **564** responsive to the reference beam **512** and the return beam **548**. The signal **564** can include an in-phase (I) component **568** and a quadrature (Q) component **572**.

[0072] The LIDAR sensor system **500** can include a receiver **576** that receives the signal **564** from the mixer **560**. The receiver **576** can generate a signal **580** responsive to the signal **564**, which can be an electronic (e.g., radio frequency) signal. The receiver **576** can include one or more photodetectors that output the signal **580** responsive to the signal **564**.

[0073] The LIDAR sensor system **500** can include a processing system **590**, which can be implemented using features of the vehicle control system **120** described with reference to FIG. **1**. The processing system **590** can process data received regarding the return beam **548**, such as the signal **580**, to determine parameters regarding the object such as range and velocity. The processing system **590** can include a scanner controller **592** that can provide scanning signals to control operation of the optics **532**, such as to control the motor **540** to cause the motor **540** to rotate the optics **532** to achieve a target scan pattern, such as a sawtooth scan pattern or step function scan pattern. The processing system **590** can include a Doppler compensator **594** that can determine the sign and size of a Doppler shift associated with processing the return beam **548** and a corrected range based thereon along with any other corrections. The processing system **590** can include a modulator controller **596** that can send one or more electrical signals to drive the modulator **514**.

[0074] The processing system **590** can include or be communicatively coupled with a vehicle controller **598** to control operation of a vehicle for which the LIDAR sensor system **500** is installed (e.g., to provide complete or semi-autonomous control of the vehicle). For example, the vehicle controller **598** can be implemented by at least one of the LIDAR sensor system **500** or control circuitry of the vehicle. The vehicle controller **598** can control operation of the vehicle responsive to at least one of a range to the object or a velocity of the object determined by the processing system **590**. For example, the vehicle controller **598** can transmit a control signal to at least one of a steering system or a braking system of the vehicle to control at least one of speed or direction of the vehicle.

3. Systems and Methods of LIDAR Sensor Systems Having Amplifier Protection Circuits

[0075] LIDAR sensor systems in accordance with the present disclosure can include a protection circuit, such as to mitigate deleterious effects on amplifiers of the LIDAR sensor used to amplify signals transmitted for determining parameters of objects in an environment around a vehicle, including but not limited to range, velocity, and/or Doppler parameters. Under various operating conditions for vehicle implementations, it can be useful for the outputted signal (e.g., beam) to be amplified and otherwise modified or controlled in a specific manner in order to achieve target performance with respect to information regarding the environment that can be extracted from a return beam from reflection of the outputted signal. The amplifier can be affected by inputs to the amplifier used to achieve the target performance for the LIDAR sensor system; systems and methods in accordance with the present disclosure can include a protection circuit in the LIDAR sensor system to address such effects. For example, the LIDAR sensor systems can include a protection circuit to detect a condition (e.g., a missing signal) associated with a modulated beam and to control input of the modulated beam to an amplifier (e.g., EDFA) and/or control the electrical input to the amplifier in response to a detection of the condition. For example, the protection circuit can evaluate at least one of the modulation signal or a parameter of the modulated beam. For example, in response to a detection of the condition, the protection circuit can control an optical attenuator, modulator, or optical amplifier to eliminate the input to the amplifier (e.g., EDFA). By controlling the optical and/or electrical input to the amplifier in response to a detection of such a condition, damages to the system (e.g., optical components) can be prevented, thereby allowing for reliable operation of the LIDAR sensor system, and thus of the autonomous vehicles having the LIDAR sensor system. For example, the protection circuit disclosed herein can provide improvements in controlling autonomous vehicles. In response to a detection of a condition (e.g., a missing signal in the LIDAR system), the protection circuit can generate an indication of the condition. The protection circuit can provide the indication to a vehicle controller, which then can operate the vehicle (e.g., controlling a steering system, a braking system, etc.) based at least on the indication of the condition. This improves stability in operating the LIDAR system, and thus the autonomous vehicles.

[0076] FIG. **7** depicts a block diagram illustrating a LIDAR sensor system **700** including a protection circuit **710** according to some implementations. The LIDAR sensor system **700** may be substantially similar to and/or incorporate features of the LIDAR sensor system **500**. For example, the LIDAR sensor system **700** includes the laser source **504**, the modulator **514**, and the amplifier **520**. The protection circuit **710** can be operably coupled to the modulator **514** and the amplifier **520**. The protection circuit **710** can detect a condition associated with optical signals (e.g., the beam **516**, a modulated beam, a modulation signal, etc.). Based on a detection of the condition, the protection circuit **710** can control an input to the amplifier **520**.

[0077] The amplifier **520** can include a high-power amplifier to amplify a light signal. For example, the amplifier **520** may be a semiconductor amplifier, an Erbium Doped Fiber Amplifier (EDFA), etc. Amplifiers (e.g., EDFAs, high-power amplifiers, etc.) can be damaged when operated at high optical power (or fluence). For example, when a pulse (e.g., a modulated beam generated by the modulator **514**, a modulation signal provided to the modulator **514**, etc.) is missing, or otherwise an issue occurs in an electro-optic component (e.g., the modulator **514** etc.), Stimulated Brillouin Scattering (SBS) could be initiated. When SBS is present, a system can be more susceptible to damages, particularly in fiber components (e.g., fiber, connectors, EDFA, isolators, etc.).

[0078] As discussed in greater detail with respect to FIG. **8**, the protection circuit **710** can provide optical and/or electrical protection of the system by detecting a condition (e.g., a missing signal) and controlling an input signal to the amplifier **520** in response to a detection of the condition. For example, the protection circuit **710** can evaluate at least one of the modulation signal or a parameter of the modulated beam. For example, in response to a detection of the condition, the protection circuit **710** can control an optical attenuator to eliminate the input to the amplifier **520**. By controlling the input to the amplifier **520** in response to a detection of such a condition, damages to the LIDAR sensor system **700** can be prevented, thereby allowing for reliable operation of the LIDAR sensor system **700**. The LIDAR sensor system **700** can be operated in combination with any of the LIDAR sensor systems and the autonomous vehicles discussed with respect to FIG. **1** to FIG. **6**. In some implementations, the LIDAR sensor system **700** can be configured for operation using a continuous wave (CW) modulation or a quasi-CW modulation. In some implementations, the LIDAR sensor system **700** can be configured for operation with any range of detecting distances (e.g., about 300 meters, 250 meters, 100 meters, 30 meters, etc.). For example, an amplified beam of the LIDAR sensor system **700** can have a parameter (e.g., a magnitude, a signal strength, a power, an amplitude, etc.) greater than a threshold used to transmit an output beam with sufficient signal strength to at least 250 meters.

[0079] FIG. **8** depicts a block diagram illustrating an example of a LIDAR sensor system **800** including the protection circuit **710** according to some implementations. The LIDAR sensor system **800** may be substantially similar to and/or incorporate features of the LIDAR sensor system **500**. For example, the LIDAR sensor system **800** as depicted in FIG. **8** includes the laser source **504**, the modulator **514**, and the amplifier **520**. The protection circuit **710** can be operably coupled to the modulator **514** and the amplifier **520**

such that a modulated beam from the modulator **514** is transmitted to the amplifier **520** through the protection circuit **710**. The protection circuit **710** can include a photodetector (PD) monitor **810**, a radio-frequency (RF) monitor **820**, a circuit control **830**, and an attenuator **840**. In some implementations, any of components in the protection circuit **710** and the LIDAR sensor system **800** can be integrated on a chip or an integrated circuit. For example, at least one of the modulator **514**, the amplifier **520**, at least one of components in the protection circuit **710** can be integrated on a chip or an integrated circuit.

[0080] The modulator **514** can receive a beam **806** from the laser source **504** and generate a modulated beam **807**. The modulator **514** can include or be connected to a direct-current (DC) input **816**, which can provide a DC bias **817** to the modulator **514**. The modulator **514** can include or be connected to a radio-frequency (RF) input **818**, which can provide an RF modulation signal **819** to the modulator **514**. Based on the DC bias **817** and the RF modulation signal **819**, the modulator **514** can generate the modulated beam **807**.

[0081] The photodiode (PD) monitor **810** can monitor the modulated beam **807**. For example, the PD monitor **810** can include one or more photodiodes (or other optical-to-electrical conversion components) that can receive the modulated beam **807** or a portion thereof and generate an electrical signal representative of the modulated beam **807** in response to receipt of the modulated beam **807**. The PD monitor **810** can be optically coupled to the modulator **514** and can receive a monitoring beam **812** of the modulated beam **807**. For example, the modulator **514** can be coupled to an optical splitter (not shown) to split the modulated beam **807** into the monitoring beam **812** and direct a remaining portion of the modulated beam **807** (e.g., portion **808**) to the attenuator **840**. In some implementations, the modulator **514** can be coupled to a photodiode to monitor the modulated beam **807**. For example, a photodiode can be optically coupled to the modulator **514** to monitor the modulated light **807** (e.g., scattering of the modulated light **807**, within a fiber or in free space) In response to receipt of the monitoring beam **812**, the PD monitor **810** can convert the monitoring beam **812** into an electrical signal **814** (hereinafter, referred to as "PD monitoring signal"). The PD monitor **810** can send the PD monitoring signal **814** to the circuit control **830** for the circuit control **830** to process the PD monitoring signal **814**. In some implementations, the PD monitor **810** and the circuit control **830** can be combined/integrated to receive the monitoring beam **812**, converting the same, and analyze the PD monitoring signal **814**. Based on the analysis, the circuit control **830** can detect a predetermined condition associated with the modulated beam **807**. In response to a detection of the predetermined condition, the circuit control **830** can control the attenuator **840** to control input of the modulated beam to the amplifier **520**. In some implementations, the attenuator **840** can be implemented using component(s) including but not limited to a modulator an optical amplifier.

[0082] The RF monitor **820** can monitor the RF modulation signal **819**. For example, the RF monitor **820** can include one or more circuit components (or components for detecting an RF signal) that can receive the RF modulation signal **819** or a portion thereof **822** (hereinafter referred to as a monitoring signal **822**) and generate an electrical signal representative of the RF modulation signal **819** in response to receipt of the monitoring signal **822**. The RF monitor **820** can be operably coupled to the RF input **818** and can receive the monitoring signal **822** of the RF modulation signal **819**. For example, the RF input **818** can be coupled to an RF splitter (not shown) to split the RF modulation signal **819** into the monitoring signal **822** and direct the remaining portion to the modulator **514**. In some implementations, the RF input **818** can be coupled to a large impedance direct connection, an RF directional coupler, etc. to provide the RF modulation signal **819** to the modulator **514** and the RF monitor **820**. In response to receipt of the monitoring signal **822**, the RF monitor **820** can send an RF monitoring signal **815** to the circuit control **830** so that the circuit control **830** analyzes the RF monitoring signal **815**. In some implementations, the RF monitor **820** and the circuit control **830** can be combined/integrated to receive the monitoring signal **822** and analyze the RF monitoring signal **815**. Based on the analysis, the circuit control **830** can detect a predetermined condition associated with the RF modulation signal **819**. In response to a detection of the predetermined condition, the circuit control **830** can control the attenuator **840** to control input of the modulated beam to the amplifier **520**.

[0083] As discussed above, the circuit control **830** can monitor and analyze the PD monitoring signal **814** and the RF monitoring signal **815** to detect a condition associated with the modulated beam **807**. In some examples, the condition may be a predetermined condition including a detection of a missing pulse in the RF modulation signal **819** and/or a missing pulse in the modulated beam **807**. The predetermined condition may be associated with any parameter of the RF modulation signal **819** and/or the modulated beam **807**. For example, the predetermined condition may be satisfied when any of physical properties (e.g., amplitude, frequency, phase, etc.) of the RF modulation signal **819** and/or the modulated beam **807** is distorted, missing, abnormal, or otherwise different from a preset value. For example, the predetermined condition may be satisfied when any of physical properties (e.g., amplitude, frequency, phase, etc.) of the RF modulation signal **819** and/or the modulated beam **807** is too high (e.g., above a maximum threshold) or too low (e.g., below a minimum threshold). The circuit control **830** can evaluate at least one parameter associated with the PD monitoring signal **814** and/or at least one parameter associated with the RF monitoring signal **815** to detect the predetermined condition. In some examples, the condition may vary depending on any of the LIDAR sensor system **800** and/or the protection circuit **710**. For example, the condition may be met when a parameter associated with the RF modulation signal **819** and/or the modulated beam **807** is below or above a threshold that varies based on settings or moving averages, etc.

[0084] In some implementations, the circuit control **830** can detect the condition is satisfied with respect to the RF input **818** and/or the RF modulation signal **819**, when the circuit control **830** detects the condition based on analysis of the RF monitoring signal **815**. In response, the circuit control **830** can control the attenuator **840** and/or the electrical input to the amplifier **520**. For example, a pulse of the RF modulation signal **819** may be missing, and the circuit control **830** can detect the missed pulse based on analysis of the RF monitoring signal **815**. The circuit control **830** can control the attenuator **840** and/or the electrical input to the amplifier **520** in response to a detection of the missed pulse, before the amplifier **520** can be damaged.

[0085] In some implementations, the circuit control **830** can detect that the condition is satisfied with respect to the modulator **514** and/or the modulated beam **807** based on analysis of the PD monitoring signal **814**. In response to detection of the condition, the circuit control **830** can control the attenuator **840** and/or the electrical input to the amplifier **520**. For example, a pulse of the modulated beam **807** may be missing, and the circuit control **830** can detect the missed pulse based on analysis of the PD monitoring signal **814**. The circuit control **830** can control the attenuator **840** and/or the electrical input to the amplifier **520** in response to a detection of the missed pulse, before the amplifier **520** can be damaged.

[0086] In some implementations, the circuit control **830** can detect that the condition is satisfied with respect to any of the RF input **818**, the DC input **816**, the RF modulation signal **819**, the modulator **514** and/or the modulated beam **807**, in response to detection of the condition based on analysis of the PD monitoring signal **814** and the RF monitoring signal **815**. In response to the detection of the condition, the circuit control **830** can control the attenuator **840** and/or the electrical input to the amplifier **520** before the amplifier **520** can be damaged.

[0087] In some implementations, the circuit control **830** can compare the PD monitoring signal **814** and the RF monitoring signal **815**. Based on a comparison, the circuit control **830** can determine that the condition is satisfied, for example in response to a mismatch between the PD monitoring signal **814** and the RF monitoring signal **815**. For example, the PD monitoring signal **814** and the RF monitoring signal **815** may be significantly different in one of parameters (e.g., phase, frequency, amplitude, etc.), and the circuit control **830** can detect the difference and determine that the condition is satisfied with respect to at least one of the modulator **514**, the RF input **818**, the DC input **816**, the modulated beam **807**, the RF modulation signal **819**, and/or the DC bias **817**. In response, the circuit control **830** can control the attenuator **840** and/or the electrical input to the amplifier **520**, before the amplifier **520** can be damaged.

[0088] The circuit control **830** can control the attenuator **840** and/or the electrical input to the amplifier **520** to control input of the modulated beam. The attenuator **840** can include an optical attenuator, a variable optical attenuator (VOA), a fast VOA, a high-speed variable fiber optical attenuator, a modulator, an optical amplifier, or any component configured to attenuate the modulated beam **808**, prevent the modulated beam **808** from entering the amplifier **520**, or otherwise configured to control/adjust/alter the modulated beam **808**, or various combinations thereof. The attenuator **840** can be optically coupled to the modulator **514** and the amplifier **520** such that the attenuator **840** can receive the modulated beam **808**, control/alter/adjust/attenuate, etc. at least one parameter (e.g., amplitude) of the modulated beam **808**, and direct the beam **809** to the amplifier **520**. In some implementations, the attenuator **840** can prevent the modulated beam **808** from entering the amplifier **520**. For example, the attenuator **840** can be turned to a maximum attenuation level to entirely eliminate the modulated beam **808** (e.g., the beam **809** not entered to the amplifier **520**). In response to a determination that the condition is not met, the attenuator **840** can simply direct the modulated beam **808** (e.g., the beam **809**) to the amplifier **520**.

[0089] In some implementations, the protection circuit **710** may include a high-power and/or high-frequency semiconductor device configured to operate under high-power and high-frequency conditions. For example, the protection circuit **710** may include but not limited to, an optical amplifier, a polarization rotator, a micro-electromechanical systems (MEMS) (e.g., a MEMS switch), an InP-based attenuator, a Mach-Zehnder switch, etc.

[0090] In some implementations, the circuit control **830** can perform an alert/control operation **850**. In response to a detection of the condition, the circuit control **830** can control the attenuator **840** and/or control the electrical input to the amplifier **520** and perform the alert/control operation **850**. The alert/control operation **850** may include generating an alert indicating the condition and sending to a user, vehicle maintenance service, LIDAR maintenance service, a computing system, etc. The alert/control operation **850** may include sending an indication of the condition to a vehicle controller and controlling, by the vehicle controller, operation of the vehicle based at least on the indication of the condition. For example, the circuit control **830** can send an alert to a vehicle controller in response to a detection of the condition, and the vehicle controller can control operation of the vehicle (e.g., a steering system, a braking system, etc.).

[0091] The protection circuit discussed herein (e.g., **710**) can provide protection of the LIDAR sensor system (e.g., **700**) and reliable operations of the LIDAR sensor system and thus of autonomous vehicles equipped with the same. More specifically, the protection circuit can provide optical and/or electrical protection of the LIDAR sensor system by detecting the condition associated with a modulated beam that may cause damages to the amplifier and/or the LIDAR sensor system and controlling one or more components to adjust and/or eliminate the modulated beam and/or control the electrical input to the amplifier **520**. In addition, the protection circuit can perform an alert/control operation (e.g., **850**) to further control the LIDAR sensor system and/or the autonomous vehicle. This prevents damages to the LIDAR sensor system while allowing for reliable operation of the autonomous vehicle as well as the LIDAR sensor system.

[0092] Referring further to FIG. **7**, the LIDAR sensor system **700** can include a power circuit **720** coupled with the amplifier **520**. The power circuit **720** can be used to provide power to the amplifier **520**, such as to activate the amplifier **520** (e.g., turn the amplifier **520** on/off). For example, the power circuit **720** can provide power to the amplifier **520** to activate the amplifier **520** in a manner that avoids or reduces the likelihood of SBS (e.g., when used in combination with control of the attenuator **840**). In some implementations, one or more components of the power circuit **720** are implemented by the protection circuit **710** (or vice versa). In some implementations, the power circuit **720** controls the power provided to the amplifier **520** to turn off the amplifier until a safe condition is achieved, which can be useful for avoiding SBS.

[0093] The power circuit **720** can include, for example, a power delivery circuit **730** coupled with a power supply **740**. The power supply **740** can include or be coupled with any of various power sources of the LIDAR sensor system **700** and/or the vehicle **100**, such as to provide a supply voltage to the power delivery circuit **730**.

[0094] The power delivery circuit **730** can control power (e.g., current and/or voltage) provided to the amplifier **520**. For example, the power delivery circuit **730** can decrease a rate of power delivery to the amplifier **520** relative to a rate of power received by the power delivery circuit **730** from the power supply **740**, such as to interrupt, smooth, and/or slow the power delivery.

[0095] In some implementations, the power delivery circuit **730** includes a capacitor, which can facilitate the smoothing of the power delivery to the amplifier **520**. In some implementations, the power delivery circuit **730** includes a resistor-capacitor (RC) circuit. For example, the power delivery circuit **730** can include a resistor and capacitor coupled with the power supply **740** (e.g., with the supply voltage from the power supply **740**) and with the amplifier **520**. At least one of the resistor can have a resistance or the capacitor can have a capacitance sized to facilitate controlling the power delivery according to the target metric for the power delivery.

[0096] The LIDAR sensor system **700** (e.g., one or more processors coupled with the power circuit **720**) can control operation of the power delivery circuit **730** and/or the attenuator **840**, such as to control a rate at which the power delivery circuit **730** activates the amplifier **520**. For example, the LIDAR sensor system **700** can cause the power delivery circuit **730** to activate the amplifier **520**, using power from the power supply **740**, based on a pulse rate for output of the amplified beam from the amplifier **520**. In some implementations, the LIDAR sensor system **700** determines the pulse rate based on a scan rate of operation of the scanning optics **532**.

[0097] FIG. **9** depicts a flow diagram showing an example method **900** for operating a LIDAR sensor system including a protection circuit according to some implementations. The method **900** is merely an example and is not intended to limit the present disclosure. Accordingly, it is understood that additional operations may be provided before, during, and after the method **900**. The method **900** can be performed with at least one of components discussed with respect to FIG. **1** to FIG. **8**. For example, the method **900** can be performed with the LIDAR sensor system **700** including the protection circuit **710**.

[0098] At **910**, source light (e.g., **806**) can be generated and directed to a modulator (e.g., **514**). The modulator can receive the source light to generate modulated light (e.g., **807**). At **920**, the modulated light can be generated by modulating the source light based on modulation signals (e.g., a DC bias **817**, a RF modulation signal **819**).

[0099] At **930**, the modulation signals and the modulated light can be monitored to detect the condition associated with the modulated light. The modulation signals can be monitored to detect whether the modulation signals satisfy the condition. The modulated light can be monitored to detect whether the modulated beam satisfies the condition. The modulated light can be monitored to detect whether the modulated light satisfies the condition.

[0100] At **940**, in response to a detection of the condition, input of the modulated light to an amplifier (e.g., **520**) can be controlled. For example, any of parameters (e.g., amplitude) of the modulated light can be adjusted in response to a detection of the condition. For example, the modulated light can be entirely eliminated thereby preventing any light from entering the amplifier, in response to a detection of the condition. In some implementations, at **940**, an alert/control operation (e.g., **850**) can be further performed in response to a detection of the condition. For example, at **940**, the LIDAR sensor system and/or the autonomous vehicle can be controlled in response to a detection of the condition.

[0101] Having now described some illustrative implementations, it is apparent that the foregoing is illustrative and not limiting, having been presented by way of example. In particular, although many of the examples presented herein involve specific combinations of method acts or system elements, those acts and those elements can be combined in other ways to accomplish the same objectives. Acts, elements and features discussed in connection with one implementation are not intended to be excluded from a similar role in other implementations or implementations.

[0102] The phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of "including" "comprising" "having" "containing" "involving" "characterized by" "characterized in that" and variations thereof herein, is meant to encompass the items listed thereafter, equivalents thereof, and additional items, as well as alternate implementations consisting of the items listed thereafter exclusively. In one implementation, the systems and methods described herein consist of one, each combination of more than one, or all of the described elements, acts, or components.

[0103] Any references to implementations or elements or acts of the systems and methods herein referred to in the singular can also embrace implementations including a plurality of these elements, and any references in plural to any implementation or element or act herein can also embrace implementations including only a single element. References in the singular or plural form are not intended to limit the presently disclosed systems or methods, their components, acts, or elements to single or plural configurations. References to any act or element being based on any information, act or element can include implementations where the act or element is based at least in part on any information, act, or element.

[0104] Any implementation disclosed herein can be combined with any other implementation or embodiment, and references to "an implementation," "some implementations," "one implementation" or the like are not necessarily mutually exclusive and are intended to indicate that a particular feature, structure, or characteristic described in connection with the implementation can be included in at least one implementation or embodiment. Such terms as used herein are not necessarily all referring to the same implementation. Any implementation can be combined with any other implementation, inclusively or exclusively, in any manner consistent with the aspects and implementations disclosed herein.

[0105] Where technical features in the drawings, detailed description or any claim are followed by reference signs, the reference signs have been included to increase the intelligibility of the drawings, detailed description, and claims. Accordingly, neither the reference signs nor their absence have any limiting effect on the scope of any claim elements.

[0106] Systems and methods described herein may be embodied in other specific forms without departing from the characteristics thereof. Further relative parallel, perpendicular, vertical or other positioning or orientation descriptions include variations within +/−10% or +/−10 degrees of pure vertical, parallel or perpendicular positioning. References to "approximately," "about" "substantially" or other terms of degree include variations of +/−10% from the given measurement, unit, or range unless explicitly indicated otherwise. Coupled elements can be electrically, mechanically, or physically coupled with one another directly or with intervening elements. Scope of the systems and methods described herein is thus indicated by the appended claims, rather than the foregoing description, and changes that come within the meaning and range of equivalency of the claims are embraced therein.

[0107] The term "coupled" and variations thereof includes the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly with or to each other, with the two members coupled with each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled with each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If "coupled" or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of "coupled" provided above is modified by the plain language meaning of the additional term (e.g., "directly coupled" means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of "coupled" provided above. Such coupling may be mechanical, electrical, or fluidic.

[0108] References to "or" can be construed as inclusive so that any terms described using "or" can indicate any of a single, more than one, and all of the described terms. A reference to "at least one of 'A' and 'B'" can include only 'A', only 'B', as well as both 'A' and 'B'. Such references used in conjunction with "comprising" or other open terminology can include additional items.

[0109] Modifications of described elements and acts such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations can occur without materially departing from the teachings and advantages of the subject matter disclosed herein. For example, elements shown as integrally formed can be constructed of multiple parts or elements, the position of elements can be reversed or otherwise varied, and the nature or number of discrete elements or positions can be altered or varied. Other substitutions, modifications, changes and omissions can also be made in the design, operating conditions and arrangement of the disclosed elements and operations without departing from the scope of the present disclosure.

[0110] References herein to the positions of elements (e.g., "top," "bottom," "above," "below") are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

What is claimed:

1. A light detection and ranging (LIDAR) sensor system for a vehicle, the LIDAR sensor system comprising:

a laser source configured to output laser light;

an amplifier configured to amplify the laser light to provide amplified light;

a circuit coupled with the amplifier, the circuit configured to provide power to the amplifier to cause the amplifier to amplify the laser light, the circuit configured to control a parameter of the power; and

one or more scanning optics configured to receive the amplified light from the amplifier and to output the amplified light.

2. The LIDAR sensor system of claim **1**, wherein the parameter that the circuit is configured to control includes at least one of an amplitude of a current of the power or a rate of change of the current.

3. The LIDAR sensor system of claim **1**, wherein the amplifier comprises an Erbium doped fiber amplifier (EDFA).

4. The LIDAR sensor system of claim **1**, wherein:

the circuit comprises at least one of a capacitor or a resistor, the at least one of the capacitor or the resistor arranged in series with a power supply; and

the at least one of the capacitor or the resistor is configured to provide the power to the amplifier from the power supply.

5. The LIDAR sensor system of claim **1**, wherein the circuit is configured to control the parameter of the power based on a pulse rate of the amplified light corresponding to a scan rate at which the one or more scanning optics output the amplified light.

6. The LIDAR sensor system of claim **1**, further comprising a modulator between the laser source and the amplifier, wherein the modulator is configured to modulate at least one of a phase or a frequency of the laser light.

7. The LIDAR sensor system of claim **1**, wherein at least one of the amplifier or the circuit is integrated on a chip.

8. The LIDAR sensor system of claim **1**, wherein the LIDAR sensor system is configured to operate using a continuous wave (CW) modulation of the laser light or a quasi-CW modulation of the laser light.

9. An autonomous vehicle control system, comprising:

a laser source configured to output a laser light;

an amplifier configured to amplify the laser light to provide an amplified light;

a circuit coupled with the amplifier, the circuit configured to provide power to the amplifier to cause the amplifier to amplify the laser light, the circuit configured to control a parameter of the power;

one or more scanning optics configured to receive the amplified light from the amplifier and to output the amplified light; and

one or more processors configured to:

determine at least one of a range to an object or a velocity of the object based on a return signal from reflection of the amplified light by the object; and

control operation of an autonomous vehicle based on the at least one of the range or the velocity.

10. The autonomous vehicle control system of claim **9**, wherein the parameter that the circuit is configured to control includes at least one of an amplitude of a current of the power or a rate of change of the current.

11. The autonomous vehicle control system of claim **9**, wherein the amplifier comprises an Erbium doped fiber amplifier (EDFA).

12. The autonomous vehicle control system of claim **9**, wherein:

the circuit comprises at least one of a capacitor or a resistor, the at least one of the capacitor or the resistor arranged in series with a power supply; and

the at least one of the capacitor or the resistor is configured to provide the power to the amplifier from the power supply.

13. The autonomous vehicle control system of claim **9**, wherein the circuit is configured to control the parameter of the power based on a pulse rate of the amplified light corresponding to a scan rate at which the one or more scanning optics output the amplified light.

14. The autonomous vehicle control system of claim **9**, further comprising a modulator between the laser source and the amplifier, wherein the modulator is configured to modulate at least one of a phase or a frequency of the laser light.

15. The autonomous vehicle control system of claim **9**, wherein at least one of the amplifier or the circuit is integrated on a chip.

16. An autonomous vehicle, comprising:

a LIDAR sensor system, comprising:

a laser source configured to output laser light;

an amplifier configured to amplify the laser light to provide amplified light;

a circuit coupled with the amplifier, the circuit configured to provide power to the amplifier to cause the amplifier to amplify the laser light, the circuit configured to control a parameter of the power;

one or more scanning optics configured to receive the amplified light from the amplifier and to output the amplified light; and

one or more processors configured to determine at least one of a range to an object or a velocity of the object based on a return signal from reflection of the amplified light by the object;

a steering system;

a braking system; and

a vehicle controller configured to control operation of at least one of the steering system or the braking system based on the at least one of the range or the velocity.

17. The autonomous vehicle of claim **16**, wherein the parameter that the circuit is configured to control includes at least one of an amplitude of a current of the power or a rate of change of the current.

18. The autonomous vehicle of claim **16**, wherein the amplifier comprises a doped fiber amplifier.

19. The autonomous vehicle of claim **16**, wherein:

the circuit comprises at least one of a capacitor or a resistor, the at least one of the capacitor or the resistor arranged in series with a power supply; and

the at least one of the capacitor or the resistor is configured to provide the power to the amplifier from the power supply.

20. The autonomous vehicle of claim **16**, wherein the circuit is configured to control the parameter of the power based on a pattern by which the one or more scanning optics control an elevation angle and an azimuth angle of the amplified light.

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